

BENCHMARK: ACCURATE — fit consistent with published values

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460696.8718 +/- 0.0019 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.158 +/- 0.015                 |
| Transit depth (Rp/Rs)^2                           | 2.5 +/- 0.46 [%]                |
| Orbital Inclination (inc)                         | 83.12 +/- 1.62                  |
| Airmass coefficient 1 (a1)                        | 0.946 +/- 0.011                 |
| Airmass coefficient 2 (a2)                        | 0.036 +/- 0.021                 |
| Scatter in the residuals of the lightcurve fit is | 1.17 %                          |
| Best Comparison Star                              | #3 - [312, 55]                  |
| Optimal Aperture                                  | 3.99                            |
| Optimal Annulus                                   | 6.71                            |
| Transit Duration (day)                            | 0.0531 +/- 0.0045               |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R fit vs published     | 0.158 ± 0.015 vs 0.1594 (deviation 0.09σ) |
| Duration fit vs published | 0.0531 d vs 0.0512 d (deviation 0.41σ)    |
| Mid-time O-C              | 3.4 min                                   |
| Depth SNR                 | 10.5                                      |
| Residual scatter          | 1.17 %                                    |

Light curve

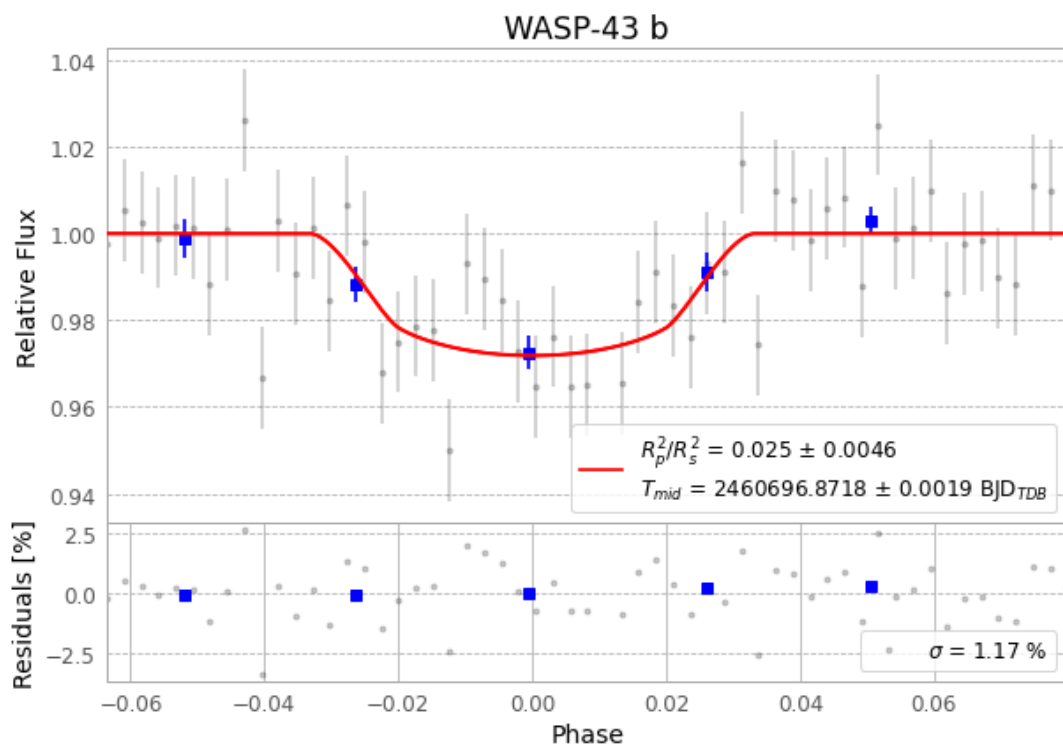


Figure 1 — FinalLightCurve\_WASP-43 b\_2025-01-21.png (SHA-256 50f6345f8ff4c712...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 225], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 71e04b31ef5b66e0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 39d9b80

## Data provenance

57 FITS frames (37 MB), WASP-43250121073315.FITS → WASP-43250121102116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-250121.log (51a209e38a87...), FinalLightCurve\_WASP-43 b\_2025-01-21.png (50f6345f8ff4...), FinalLightCurve\_WASP-43 b\_2025-01-21.csv (e065131b946b...)

## Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 09:05Z.